

Question:

A 10-month-old boy is brought to the emergency department by his parents because he has a high fever and severe cough. His fever started 2 days ago and his parents are concerned as he is now listless and fatigued. He had a similar presentation 5 months ago and was diagnosed with pneumonia caused by Staphylococcus aureus. He has been experiencing intermittent diarrhea and skin abscesses since birth. The child had an uneventful birth and the child is otherwise developmentally normal. Analysis of this patient's sputum reveals acute angle branching fungi and a throat swab reveals a white plaque with germ tube forming yeast. Which of the following is most likely to be abnormal in this patient? Answer Choices: A. Autoimmune regulator function B. Lysosomal trafficking C. NADPH oxidase activity D. Thymus development

Reasoning + Answer:

<think>The patient is a 10-month-old with a history of recurrent infections, including pneumonia caused by Staphylococcus aureus, and symptoms of fever, cough, and fatigue. He also has intermittent diarrhea and skin abscesses. His sputum and throat swab analysis reveal acute angle branching fungi and yeast, respectively. This presentation is indicative of a primary immunodeficiency, specifically a disorder of the Phagocyte NADPH Oxidase. The NADPH oxidase is a key component of the phagocyte NADPH oxidase complex, which generates reactive oxygen species necessary for the killing of certain pathogens, including fungi. The patient's history of recurrent infections, the presence of acute angle branching fungi and yeast in the sputum and throat swab, and his symptoms are all consistent with a defect in the NADPH oxidase activity. </think> <answer> C. NADPH oxidase activity </answer>

